

Scores are capped at 100 points.

1. (20 points) In this problem, we will show that some variants of SAT are **NP**-complete.

- (a) (10 points) For fixed integer $k \geq 3$, show the following language is **NP**-complete:

$$\text{EXACT-}k\text{SAT} = \{\langle \phi \rangle \mid \phi \text{ is a satisfiable CNF and every clause has exactly } k \text{ literals}\}$$

- (b) (10 points) For fixed integer $k \geq 1$, show the following language is **NP**-complete:

$$k\text{MANY-SAT} = \{\langle \phi \rangle \mid \text{the number of satisfying assignments of } \phi \geq k\}$$

2. (20 points) The class **DP** is defined as the set of languages L for which there are two languages $L_1 \in \mathbf{NP}$, $L_2 \in \mathbf{coNP}$ such that $L = L_1 \cap L_2$. Show that:

- (a) (10 points) $\text{EXACT-INDSET} \in \mathbf{DP}$, where:

$$\text{EXACT-INDSET} = \{\langle G, k \rangle \mid G \text{ has an independent set of size exactly } k\}$$

- (b) (10 points) Every language in **DP** is polynomial-time reducible to EXACT-INDSET.

Hint: Do not confuse **DP** with $\mathbf{NP} \cap \mathbf{coNP}$, which may seem superficially similar.

3. (20 points) Prove that the following languages are **PSPACE**-complete:

- (a) (10 points) $\text{SPACE-TMSAT} = \{\langle M, w, 1^n \rangle : \text{DTM } M \text{ accepts } w \text{ in space } n\};$

- (b) (10 points) $\text{LINEAR-SPACE-TMSAT} = \{\langle M, w \rangle : \text{DTM } M \text{ accepts } w \text{ in space } |w|+1\}.$

4. (20 points) In this problem, we will talk about some modifications of the definition of **NL** and **NL**-completeness, and see how they change the class being defined.

- (a) (10 points) In the certificate definition of **NL**, we only allow the verifier machine to read the certificate from left to right. If we allow the verifier machine to move its head back and forth, what class of languages do we get? Prove your answer.

Hint: The answer can be given in terms of a well-known complexity class.

- (b) (10 points) We know that **NL**-completeness is defined with respect to log-space reductions $\leq_l$. If we use polynomial-time reductions $\leq_p$ instead, give the set of all '**NL**-complete' languages and prove your answer.

Hint: The answer can be given by removing some 'non-complete' languages from **NL**.

5. (30 points) Prove that the following languages are **NL**-complete:

- (a) (10 points) $A_{\text{NFA}} = \{\langle N, w \rangle \mid N \text{ is an NFA that accepts } w\};$

- (b) (10 points) $B_{\text{NFA}} = \{\langle N, M \rangle \mid N, M \text{ are NFAs and } L(N) \cap L(M) \neq \emptyset\};$

- (c) (10 points) 2SAT.